

Daniel Rocha

Senior Data Engineer

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SUMMARY

Senior Data & Analytics Engineer Consultant with 10+ years in data engineering, specializing in Microsoft Azure platforms including Azure Data Factory, Azure Databricks, Synapse Analytics, ADLS, and Azure SQL Database. Proven at designing lakehouse architectures, metadata-driven data orchestration, and complex SQL and dimensional models / star schemas that enable BI, analytics, and AI/ML. Mentored teams and led migrations that realized 3-4x throughput and reduced production incidents by 50-60% while delivering cost optimizations. Passionate about problem solving, relationship-building, Adaptability/flexibility, and fostering a culture of continuous learning and growth.

SKILLS

- Azure Data Platform (Azure Data Factory, ADLS, Synapse Analytics, Azure SQL Database, Fabric)
- Azure Databricks & Spark (Delta Lake, SparkSQL, notebook CI/CD, performance tuning, partitioning)
- Data Orchestration & Pipelines (metadata-driven patterns, Azure Data Factory, Apache Airflow, orchestration design)
- Dimensional Modeling & Star Schemas (Kimball-style modeling, conformed dimensions, slowly changing dimensions)
- API Ingestion & JSON Handling (OAuth 2, REST APIs, JSON flattening/normalization to SQL structures)
- Data Lakehouse & Warehousing (Delta Lake, Snowflake, data lakes + data warehouses hybrid architectures)
- Data Quality, Observability & Lineage (automated validation checks, monitoring, schema evolution, lineage documentation)
- SQL Mastery (complex SQL, T-SQL, query optimization, Azure SQL Database, analytical SQL patterns)
- Streaming & Near-Real-Time (Kafka, Spark Streaming, near-real-time pipelines for operational dashboards)
- CI/CD & DevOps for Data (Git, Azure DevOps, notebook testing, release pipelines, deployment governance)
- Security & Compliance (role-based access control, GDPR/HIPAA-aware data handling, encryption patterns)
- Leadership & Mentoring (technical mentorship, workshops, training, documentation, diagramming champion)
- Soft Skills: Curiosity, Emotional intelligence, Persistence, Creativity, Resourcefulness, Relationship-building

WORK HISTORY

Senior Data Engineer

Oracle | *Fort Worth, TX (remote)*

September 2020 - Current

- Spearheaded enterprise lakehouse initiatives combining Azure Databricks, Delta Lake, ADLS, and Snowflake to deliver governed analytics platforms supporting BI and AI workloads.
- Architected metadata-driven ingestion and orchestration frameworks using Azure Data Factory, Apache Airflow, Python, and Spark to process hundreds of GB to multiple TB per day from APIs, CDC streams, relational systems, and vendors.
- Designed dimensional models and star schemas (Bronze/Silver/Gold medallion patterns) with clear ownership, enabling self-service BI and reducing report build time by 40%.
- Partnered with analytics, finance, product, and ML teams to translate ambiguous business requirements into scalable data products with SLAs, runbooks, and onboarding documentation.
- Implemented API ingestion patterns including OAuth 2 authentication and JSON flattening/normalization to relational structures for downstream analytics.
- Established data quality, observability, and lineage standards with automated validation checks and anomaly detection, lowering production incidents by 50–60%.
- Built CI/CD pipelines for notebooks, dbt models, and SQL artifacts using Git and Azure DevOps to standardize code reviews, testing, and releases across environments.
- Optimized Spark workloads through partitioning strategies, adaptive execution, predicate pushdown, and cluster sizing, reducing cloud compute spend by 25–30%.
- Mentored senior and mid-level engineers via architecture reviews, pair programming, and runbook authoring to elevate operational maturity and on-call readiness.
- Orchestrated real-time and near-real-time pipelines with Kafka and Spark Streaming to support operational dashboards and alerting use cases.
- Modernized legacy ETL to Spark-based ELT pipelines and cloud storage, realizing 3–4x throughput improvements and more predictable SLA attainment.
- Documented architecture diagrams, data dictionaries, and technical runbooks; served as documentation champion during client workshops and training sessions.

Senior Data Engineer

Chime | *Salt Lake City, UT*

September 2017 - April 2020

- Created and maintained enterprise ELT pipelines using Talend, Spark, Hadoop, Python, and SQL that later migrated to Snowflake-based warehouses.
- Migrated on-prem and legacy warehouse systems to cloud-first architectures, laying the foundation for Snowflake adoption and modern ELT patterns.
- Developed Python-based transformation and validation frameworks to standardize data quality prior to loading analytics stores.
- Modeled analytical datasets and reporting layers aligned to dimensional modeling best practices to support scalable BI and ad-hoc analysis.
- Delivered REST APIs (Flask/FastAPI) to support ingestion, reconciliation, and analytics services with robust error handling and monitoring.

- Deployed dashboards in Power BI and Tableau backed by optimized warehouse models to accelerate decision-making across finance, operations, and product.
- Automated monitoring and alerting using Elasticsearch and Kibana to ensure pipeline reliability and SLA compliance.
- Collaborated with product, QA, and operations in Agile environments, facilitating technical design reviews and production readiness.

Data Engineer

CloudBees | *Arlington, VA*

July 2014 - July 2017

- Constructed foundational ETL pipelines and analytical reporting workflows using Python and SQL to enable business intelligence use cases.
- Developed high-throughput ingestion services and APIs to transform transactional data into analytics-ready stores.
- Automated infrastructure provisioning and deployments using AWS CloudFormation and Elastic Beanstalk to improve reliability and deployment speed.
- Authored reusable data models, documentation, and onboarding materials that supported future cloud warehouse migrations to Snowflake.
- Streamlined data reconciliation processes with reconciliation services and reconciliation dashboards to ensure data accuracy for stakeholders.
- Secured data transfer and storage patterns aligned with company policies to support compliance and controlled access.

EDUCATION

Bachelor's Degree, Computer Science @ California State University

2013